

PROBLEM SET 2

Due date: March 29, 2013 at the recitation session

1. Prove that if a 2×2 matrix has a *strict* Nash equilibrium that is also unique, then the game can be solved via iterated strict dominance.
2. Consider the game depicted in table 1 where $a_i > 0$ and $b_i \in \mathbb{R}$ for $i = 1, 2$.

		Pl 2	
		H	T
Pl 1	H	$a_1 + b_1, -a_2 + b_2$	$-a_1 + b_1, a_2 + b_2$
	T	$-a_1 + b_1, a_2 + b_2$	$a_1 + b_1, -a_2 + b_2$

Figure 1

- (a) Verify that no pure-strategy NE of this game exists.
 - (b) Compute the mixed-strategy NE, $(\sigma_1^*(H), \sigma_2^*(H))$ of this game.
 - (c) Explain briefly (in economic terms) why the NE does not depend on the parameters a_1, b_1, a_2 , or b_2 .
3. Two students ($i = 1, 2$), each having an endowment of time $T_i = 1$, are working on a project together. Each student i has to decide how to allocate his time endowment between working (t_i) and leisure (l_i). The grade each receives on the project depends on how much time they spend collectively working on it. Let $G = 50(t_1 + t_2)$ denote the grade earned on the project. The utility function of student i is given by

$$u_i(G, l_i) = a_i \sqrt{2G} + l_i$$

The students simultaneously choose how much time they will allocate to the project.

- (a) Set up the optimization problem for player i and derive the best response function $t_i^{br}(t_j)$.
 - (b) Suppose that $\frac{1}{10} > a_1 > a_2$. Carefully draw the best response functions with t_1 on the x -axis. Characterize all Nash equilibria (if any) in pure strategies.
 - (c) Suppose that $a_1 = a_2 = a < \frac{1}{10}$. Characterize all Nash equilibria (if any) in pure strategies.
4. Mas-Colell 8.D.3